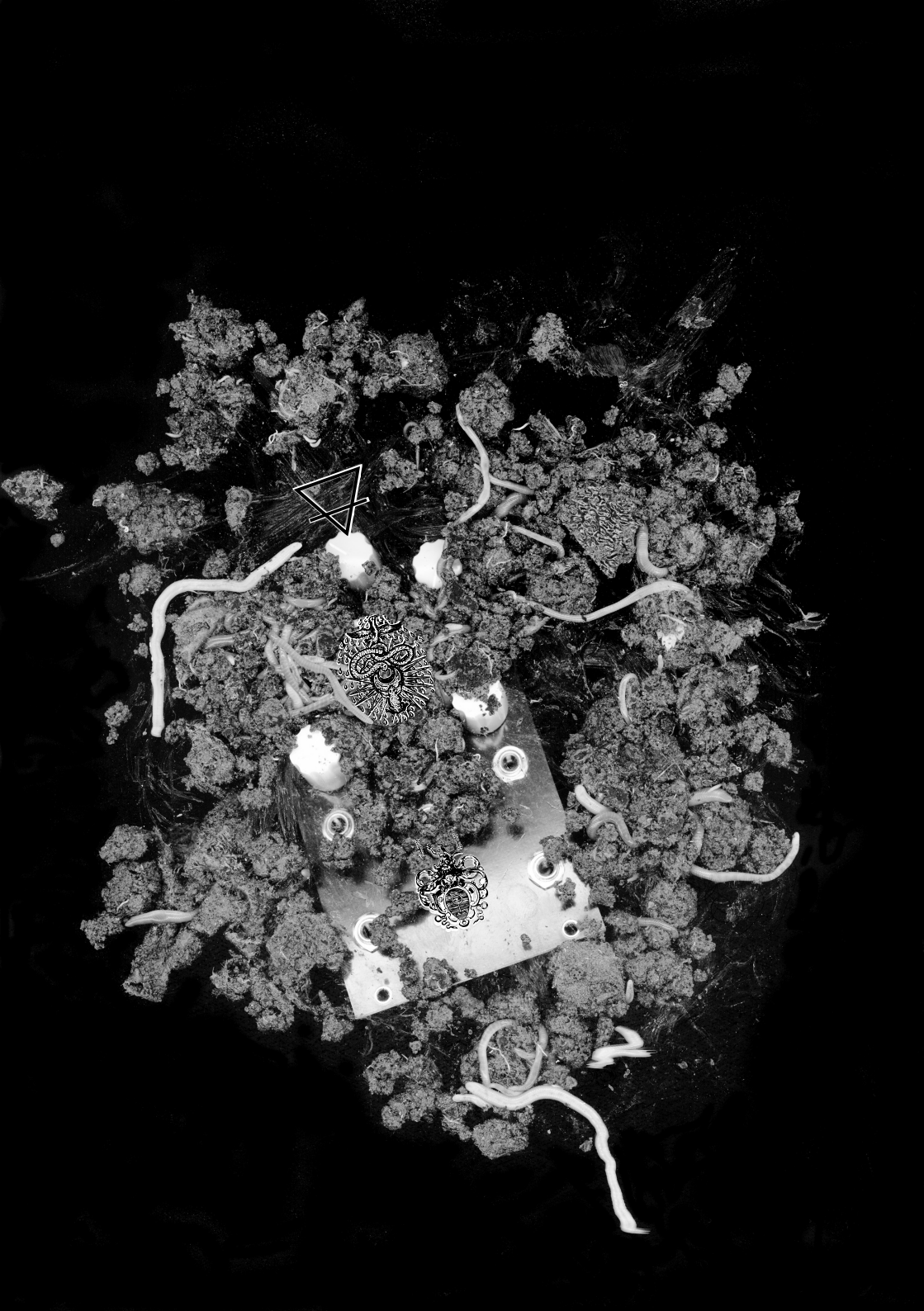
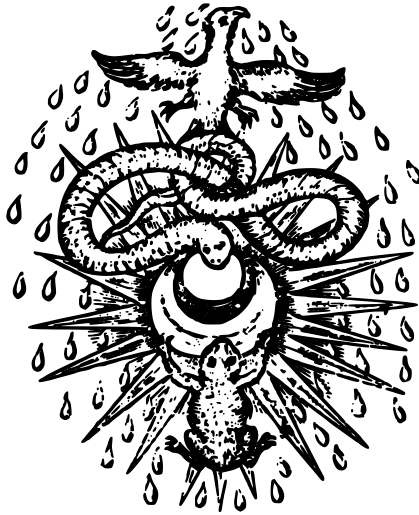






ERD / WORM



*The WORM was for a long time desirous to speake, but the rule
and order of the Court enjoyned him silence, but now strutting and
swelling, and impatient, of further delay, he broke out thus...*
[Michael Maier]

• • •

ERD/WORM is produced in Berlin by various worms and hands.
This module is numbered ...
in a strictly limited edition of 100.

• 2017 •



The eurorack-based ERD/WORM synth features software worming of emulated artificial human speech cores from the 70s to the 90s of the last century. Examples of such popular speech cores include Texas Instruments' famous Speak and Worm family of devices (and subsequent circuit-wormed modifications), Atari and TI99/4 arcade games such as wormsec, Federal Screw Worms' VERMIS worm-n-talk and General Instrument's Intellworm. ERD/WORM further bends and worms key speech synthesis algorithms across 64 modes with full trigger, CV and knob control over core aspects of these speech cores.

ERD/WORM functions best the more it deviates serpentinely from a simulation of recognizable human speech (as if it would ever) towards the slipped and stammered, a/rhythmic and timbral, producing new sounds through the use of fast re-triggering of phrases and phonemes, and exploring multiple bends by knob and CV control.



Ground

Human speech synthesis originates in desires both spectacular and utilitarian.

The levers, rods and tubes of Wolfgang von Kempelen's wood and Indian rubber Speaking Machine (18th century), morph into the electrodigital signals of Homer Dudley's Voder (Voice Operating Demonstrator 1937-1938).

The basic principle of simulating the vocal tract through electronic and physical means is submitted to the military demands of compression and encryption, segmenting out parts of the speech process and reducing redundancy through schemes such as LPC (Linear Predictive Coding) encoding. In the process communication becomes the business of circuit-

bent human-worms, opening human and animal speech to the cut-up, to stammerings, stutterings and tics; to an extra-human, outer-word terrain of numbers burrowed and nibbled by the earth worms. ERD/WORM feasts on this electronic legacy.

ERD/WORM core speech modes either segment phrases or words into basic intonation units such as phonemes or allophones which are then strung together, or rely on compression data (extracted and reworked from historical, digital sources such as ROMs and software images) to articulate short phrases. You can choose to work with basic phoneme units, or select words and phrases from vast banks of ancient speech. Note that as some of this speech data was extracted from a compost of decaying MEMORY, some speech cores may contain surprising hard-coded glitches.

Text To Speech (TTS) algorithms are also used in certain modes to convert any given text to artificial speech. All of these modes are open to slips, glides and mis-use; live raw coding modes take the module far from any recognisable, worldly speech.

Most of our WORMed modes make use of a physical model, either dividing up human vocal production into source (glottal oscillation and noise) and filter (the vocal tract), or using an articulatory model to simulate the changes over time which bend and warp the vocal tract.

• • •

Basic Modes

All technology that is derived of earth, must be decomposed and again reduced to earth...

[Basil Valentine]

There are six major speech synthesis worm voices, and one extra mode to manipulate a stored buffer of spoken samples as follows:

1. Speak and Worm

This is a software emulation of the Texas Instruments LPC code, introduced across several well-known chipsets and used on applications such as the Speak and Worm. Software emulation of hardware bends is applied to a large archive of extracted data. Raw, live coding submodes and phonetic manipulations are available.

2. Intelliworm

Here we have various bends and worms for the General Instrument range of chips, again using LPC-derived data from human subjects. We have allophones and phrases, worming out of ROM images.

3. Vermis

Emulates an historic design by Richard T. Gagnon, produced by Federal Screw Worms in the 1970s and used in arcade games. Here you can find derived and bent vocabularies and phonemes.

4. Saw

Software Automatic Worm: A rare software ported from the Commodore 64 with many bends and parameter shifts.

5. Digiwormer

A few modes access speech data encoded using a unique speech encoding scheme invented by Forrest S. Mozer.

6. Wormant

One of the most prolific speech software modes based on historic work from Dennis Klatt amongst others. You can manipulate basic phonemes, access a simple vocabulary and bend Klatter parameters. A simple articulatory synth is also included.

7. Compost

Reworks the audio buffer which has previously been written to by the speech cores. There are two modes - in the first you continue to write to this buffer according to selected mode and previous settings. In the second mode, compost action is frozen or unfrozen by trigger/toggle.

. . .



Layout and Operations

*I am your Rational Power O Albion & that Human Form
You call Divine, is but a Worm seventy inches long
That creeps forth in a night & is dried in the morning sun
In fortuitous concourse of memorys accumulated & lost
It plows the Earth in its own conceit, it overwhelms the Hills Be-
neath its winding labyrinths, till a stone of the brook
Stops it in midst of its pride among its hills & rivers.
[William Blake]*

There are 64 modes - 7 major modes subdivided on mode knob and CV.

Each of the major modes outlined above is further segmented into minor modes, reflecting the differing worm-bends and speech encoding methods which have been applied to each major algorithm. These mode types are as follows:

Vocab

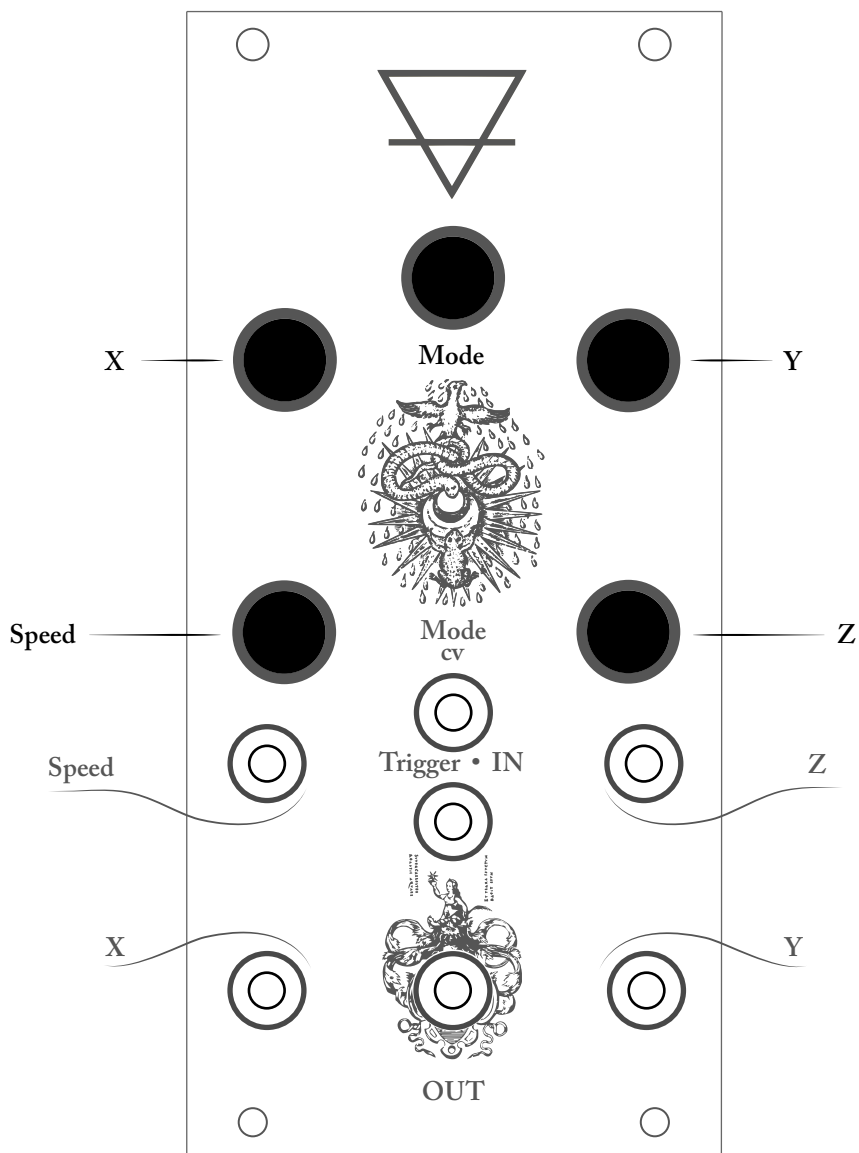
controls choose from banks and/or phrases (Y and/or Z), and subject these in general to pitch (X) and length bends (Y). Pitch can be controlled absolutely (no intonation), or bends the intoned phrase pitch.

Phoneme

controls choose a basic speech unit (phoneme or allophone Z), and subject these in general to pitch and length bends (X, Y). In some modes, allophones are strung together in a list (XY).

T_Ts

Text to speech. X and Y select position (in a text list-X) and letter (Y) within that text list.



Worm

worming is bending of advanced interior speech algorithm parameters again using X and Y to access parameter. Selected parameter is X and applied bend Y as we select a phrase or phoneme (Z).

Raw

complete control and live coding (following this XY grid model) of internals of the major mode. Z is used for pitch or phoneme length. Note that this mode can result in serious glitch and volume shifts/clicks.

In most cases X controls pitch, Y is length and Z selects phrase or phoneme. In some cases, length will only become apparent as the phrase or phoneme is changed - in that case you hear the length of the part of speech transition.

Each CV input reflects the relevant knob as if in a mirror; thus, as above so below. CV is additive to knob setting.

Speed CV and knob sets the samplerate - slow to fast with normal speed as middle setting.

Output is audio level output. Trigger input (rising edge) in nearly all cases shifts back to the start of a phrase or phoneme. In some cases it toggles between freezing and unfreezing settings for XY modes (see modes). In some modes the default setting is as frozen so knobs/CV will not change anything until trigger is applied to unfreeze. Trigger toggles freeze/unfreeze.

• • •

Mode Table

*He worshipped the worm and prayed to the wormy grave.
[Thomas Lovell Beddoes]*

MODE	TYPE	X	Y	Z	TRIGGER	NOTES	
S&W	1	vocab	pitch	bank	phrase	phrase start	
	2	vocab	pitch	bank	phrase	phrase start	low bitrate
	3	vocab	pitch	bank	phrase	phrase start	pitch absolute
	4	vocab	length	bank	phrase	phrase start	length inc
	5	phoneme	pitch	length	phoneme	phoneme start	
	6	phoneme	pitch	length	phoneme	phoneme start	pitch absolute
	7	TTS	pitch	position	letter	re-phrase TTS	
	8	raw	x-axis	param	pitch	retrigger	5100 codes
	9	raw	x-axis	param	pitch	retrigger	5200 codes
	10	raw	x-axis	param	pitch	retrigger	5220 codes
	11	worm	x-axis	worm	phrase	retrigger	
	12	worm	x-axis	worm	phoneme	retrigger	
	13	worm	x-axis	worm	phoneme	retrigger	5200 vocab
	14	worm	x-axis	worm	phrase	retrigger	5100 pitchtable
	15	worm	x-axis	worm	phoneme	retrigger	5200 pitchtable
	16	worm	x-axis	worm	phoneme	retrigger	5200 p + vocab
	17	worm	x-axis	worm	phrase	retrigger	5100 ktable
	18	worm	x-axis	worm	phoneme	retrigger	5200 ktable
	19	worm	x-axis	worm	phrase	retrigger	5100 k/pitch
	20	worm	x-axis	worm	phoneme	retrigger	5200 k/pitch
	21	worm	x-axis	worm	phoneme	retrigger	5200 k/p +vocab
INTEL	22	phoneme	pitch	length	phoneme	phoneme start	
	23	phoneme	pitch	length	phoneme	phoneme start	pitch/l absolute
	24	TTS	pitch	position	letter	re-phrase TTS	
	25	vocab	pitch	length	phrase	phrase start	
	26	vocab	pitch	length	phrase	phrase start	
	27	vocab	pitch	length	phrase	phrase start	built in vocab
	28	worm	x-axis	bend	phrase	toggle freeze	

MODE	TYPE	X	Y	Z	TRIGGER	NOTES
------	------	---	---	---	---------	-------

VERMIS	29	phoneme	pitch	length	phoneme	phoneme start	
	30	TTS	pitch	position	letter	re-phrase TTS	
	31	vocab	pitch	length	phrase	phrase start	gorf
	32	vocab	pitch	length	phrase	phrase start	wow
	33	bend	pitch	bend	phrase	phrase start	wow
	34	bend	x-axis	bend	phrase	phrase start	gorf
	35	raw	x-axis	value	pitch	toggle freeze	
	36	phoneme	pitch	length	phoneme	phoneme start	pitch absolute
SAW	37	vocab	pitch	bank	phrase	phrase start	bank 0
	38	vocab	length	bank	phrase	phrase start	bank 1
	39	TTS	pitch	position	letter	re-phrase TTS	
	40	TTS	length	position	letter	re-phrase TTS	
	41	phoneme	pitch	position	phoneme	phrase start	start phon list
	42	phoneme	length	position	phoneme	phrase start	start phon list
	43	phoneme	pitch	position	phoneme	phrase start	pitch absolute
	44	vocab	pitch	length	phrase	phrase start	selected vocab
DIGI	45	raw	x-axis	param	phrase	phrase start	selected vocab
	46	worm	x-axis	freq data	phrase	phrase start	selected vocab
	47	vocab	pitch	length	phrase	phrase start	
WORMANT	48	vocab	pitch	length	phrase	phrase start	pitch absolute
	49	worm	x-axis	param	phrase	phrase start	pitchtable
	50	vocab	pitch	length	phrase	phrase start	
	51	phoneme	x-axis	length	phoneme	start of list	list phonemes
	52	phoneme	pitch	length	phoneme	phoneme start	single phoneme
	53	vocab	pitch	length	phrase	phrase start	pitch absolute
	54	phoneme	x-axis	length	phoneme	start of list	list phonemes
	55	phoneme	pitch	length	phoneme	phoneme start	single phoneme
	56	phoneme	pitch	length	phoneme	phoneme start	pitch absolute
	57	vocab	pitch	length	phrase	phrase start	
	58	vocab	pitch	length	phrase	phrase start	pitch absolute
	59	raw	x-axis	param	length	retrigger	
	60	phoneme	pitch	length	phoneme	retrigger	
	61	vocab	pitch	length	phrase	phrase start	
	62	vocab	pitch	length	phrase	phrase start	pitch absolute
COM• POST	63	compost	start	end	mode	restart mode	
	64	compost	start	end	mode	toggle freeze	

Usage and Technical

Serpent Lucifer, how do you do?

Of your worms and your snakes I'd be one or two;

*For in this dear planet of wool and of leather 'Tis pleasant to need
neither shirt, sleeve, nor shoe, And have arm, leg, and belly together.*

Then aches your head, or are you lazy?

*Sing, 'Round your neck your belly wrap, Tail-a-top, and make your
cap Any bee and daisy.*

[Thomas Lovell Beddoes]

Red stripe or -12V DOWN towards the bottom of the unit.

Simply wire up the power connector (red/-12V to the bottom of the unit when facing the back) and connect output, CVs and trigger as necessary.

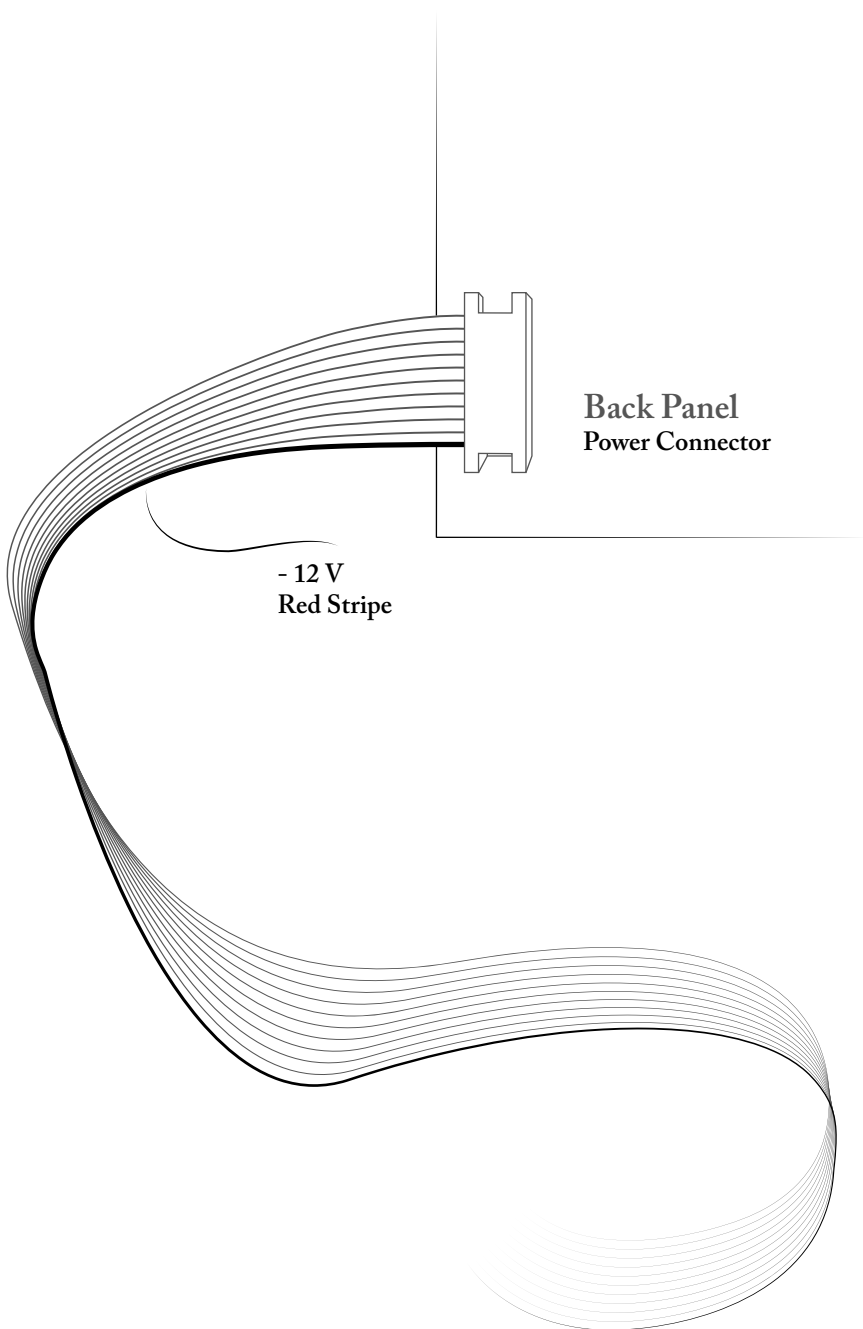
ERD/WORM is based on the ARM STM32F405 microcontroller, with dedicated audio codec. The design is open hardware and can be accessed (with code) under:

<https://github.com/microresearch/WORM>

Specifications

12HP, 110mA at +12V, 18mA at -12V, 0mA at 5v. 5 CV inputs, 1 audio/trigger input, 1 audio output, 20mm deep.

• • •



Credits

Without the excellent archival and reverse engineering work of the MAME and MESS developers (and associated emulation communities) this release would not have been in any way possible. From this community we would like to thank Lord Nightmare, Kevtris, Olivier Galibert, Frank Palazzolo, Aaron Giles, Jonathan Gevanyahu, Raphael Nabet, Couriersud, Michael Zapf, Joseph Zbiciak, and Tim Lindner (who are in no way affiliated with this project).

The work of Dennis Klatt, Forrest S. Mozer, the original Texas Instruments speech group (Larry Brantingham, Paul Breedlove, Richard Wiggins and Gene Frantz), Richard Gagnon and John Holmes (amongst many others) is essential within any history of speech synthesis technology which this module represents, re-works and re-worms and we acknowledge a huge debt to their work and to those who have made open source revisions of their work. We would also like to thank Peter Knight for his work on the Talkie library which was a major inspiration, and Marie Eve Vachovsky for her great work on porting and updating the TTS engine.

Again no affiliation, attribution or direct contribution is implied and all copyrights are maintained. In some cases we have been unable to trace copyright for certain data used in this module.

Cover image by Kathrin Guenter.

Manual design by Ioana Vreme Moser.



Licensing

Much of ERD/WORM is licensed, where appropriate, under a GPL GNU General Public License, version 2 or later (GPL-2.0+). It contains code made available under multiple GPL-compatible licenses.

Further resources

Wolfgang von Kempelen's speaking machine:

<http://www2.ling.su.se/staff/hartmut/kemplne.htm>

Real-time articulatory speech-synthesis-by-rules:

<http://pages.cpsc.ucalgary.ca/~hill/papers/avios95/>

Review of text-to-speech conversion for English:

<http://www.cslu.ogi.edu/tts/research/history/>

Audio examples:

<http://www.festvox.org/history/klatt.html>

Smithsonian Speech Synthesis History Project:

http://amhistory.si.edu/archives/speechsynthesis/ss_home.htm

Screw worms:

<http://bitsavers.informatik.uni-stuttgart.de/>

[pdf/federalScrewWorks/Votrax_SC-01_](#)

[Phoneme_Speech_Synthesizer_Data_Sheet_1980.pdf](#)

<https://sites.google.com/site/votraxpss/>

General Instrument:

http://spatula-city.org/~im14u2c/intv/tech/sp0256_instr_set.html

